

Constituency Decision Intelligence OS

An Explainable AI-driven Digital Twin for MPs and District Administrators
to optimize budget allocations · audit grievances · visualize micro-GIS asset telemetry

Hackathon Track	People's Priorities — AI for Constituency Development Planning
Solution Name	LokNiti AI — Constituency Decision Intelligence OS
Geographic Focus	West Bengal, India (4 Lok Sabha Constituency Presets)
Technology Stack	React 18 · TypeScript · Vite · Vanilla CSS · SVG GIS
GitHub Repository	https://github.com/SubhrodipPal/LokNiti-AI
Author	Subhrodip Pal

4	4	1	5+
WB Constituencies	Stakeholder Roles	Central Undo Ledger	GIS Map Layers

■ Table of Contents

1. Problem Statement	3
2. Introduction & Background	4
3. Existing Problems in Constituency Governance	5
4. Proposed Solution — LokNiti AI	6
5. System Architecture	7
6. Feature Breakdown & Methodology	8
7. West Bengal Data Profiles	11
8. Technology Stack	12
9. Advantages & Impact	13
10. Screenshots & Demo Evidence	14
11. Conclusion & Future Roadmap	16

■ 1. Problem Statement

Track: People's Priorities — AI for Constituency Development Planning

India's democratic governance structure creates an enormous challenge: over 543 Lok Sabha Members of Parliament (MPs) each represent an average of 1.5 million constituents spread across multiple assembly segments, villages, wards, and blocks. Under the MPLADS (Members of Parliament Local Area Development Scheme), each MP receives ₹5 Crore annually — ₹25 Crore over a 5-year term — to fund local infrastructure development. The central question this hackathon poses is:

"How can AI help MPs and administrators identify the highest-impact development projects, distribute constituency budgets optimally, and systematically respond to citizen grievances — especially in underserved, linguistically diverse, and geographically challenging regions?"

LokNiti AI directly addresses this challenge by building a full-stack AI governance platform that serves all four key stakeholders in the constituency governance chain: citizens, MPs, district officers, and system administrators.

■ 2. Introduction & Background

India manages one of the most complex democratic administrations in the world. Despite robust planning institutions, a persistent information asymmetry exists between citizen needs at the ward/village level and decision-making happening at the constituency or district level. MPs and officers frequently rely on manual reports, political perception, and informal channels to identify where development funds are most needed.

West Bengal — with its extraordinary geographic and cultural diversity — presents an ideal lens to study this problem. The state encompasses dense urban metros (Kolkata), dry dryland plateaus (Purulia), the Himalayan foothills (Darjeeling), and the world's largest mangrove delta (Sundarbans). Each presents a radically different infrastructure challenge, population profile, and development need.

MPLADS Scheme Background

The Members of Parliament Local Area Development Scheme (MPLADS) was introduced in 1993 to enable MPs to suggest works of capital nature based on locally felt needs, with an emphasis on the creation of durable community assets. Key parameters:

MPLADS Annual Entitlement	■5 Crore per MP per year
5-Year Term Total	■25 Crore per constituency
Fund Release	Directly to District Authority (DM/DEO)
Eligible Projects	Roads, Schools, Drinking Water, Health, Sanitation
Project Execution	State Government / Local Bodies
Current Challenge	Manual prioritization, no real-time grievance tracking

LokNiti AI modernizes this workflow end-to-end using explainable AI, multilingual NLP, interactive GIS mapping, and real-time budget simulation — transforming a manual, politically-driven funding allocation into a transparent, data-driven, citizen-first governance model.

3. Existing Problems in Constituency Governance

■ No Centralized Grievance Intelligence	Citizen complaints arrive through disparate channels — physical letters, phone calls, social media, and panchayat meetings. There is no unified system to aggregate, de-duplicate, geo-tag, and prioritize these grievances.
■ Subjective & Politically Biased Fund Allocation	Without data-driven insights, MPLADS funds are often allocated based on political considerations rather than actual infrastructure deficits. High-need remote villages consistently lose out to politically visible urban centers.
■ No Real-Time Development Index Monitoring	MPs lack a live dashboard showing the development status (roads, schools, health centers, water access) of each assembly segment. Decision-making happens with stale, annual survey data.
■ Language Barrier in Citizen Engagement	In West Bengal, citizens speak Bengali, Nepali, Santali, and several tribal languages. Government portals are almost exclusively in English or Hindi, creating a massive accessibility gap.
■ No Budget Simulation or "What-If" Planning	Officers cannot simulate the projected development impact of different fund distribution scenarios before committing. Once a sanction is issued, reversal is complex and bureaucratically slow.
■ Lack of Accountability & Audit Trail	Actions taken (or not taken) on citizen requests rarely have transparent audit logs. Citizens have no way to track whether their grievance influenced any official decision.

■ 4. Proposed Solution — LokNiti AI

LokNiti AI is a four-role, multi-panel AI governance platform designed to comprehensively solve each of the problems identified above. It creates a seamless data flow from citizen grievance submission all the way through AI-driven prioritization, MP sanction approval, officer-level execution tracking, and budget impact simulation.

■ CITIZEN PORTAL	Voice-to-text multilingual grievance submission in Bengali, Hindi, and Kannada. OCR-powered photo damage assessment. Real-time request tracking and community consensus voting.
■ MP / MINISTER DASHBOARD	Interactive GIS constituency twin with heatmaps and development index visualization. Explainable AI priority ranking with reasoning breakdown. Master budget planner with AI distribution. Complete action ledger with one-click undo.
■ OFFICER PORTAL	Project workflow management across stages (Recommended → Approved → Tender → In Progress → Completed). AI-generated warning indicators for cost overruns. Action ledger with rollback capability.
■ ADMIN CONSOLE	System weight calibration — tune the AI scoring weights for citizen demand, infrastructure gap, population density, vulnerability ratio, and more. Model activity log viewer.

■ 5. System Architecture

LokNiti AI follows a component-driven reactive architecture. The central App.tsx manages a global state store which synchronizes all four role dashboards. Database operations are simulated through a realistic mock database (mockDb.ts) containing real West Bengal constituency data.

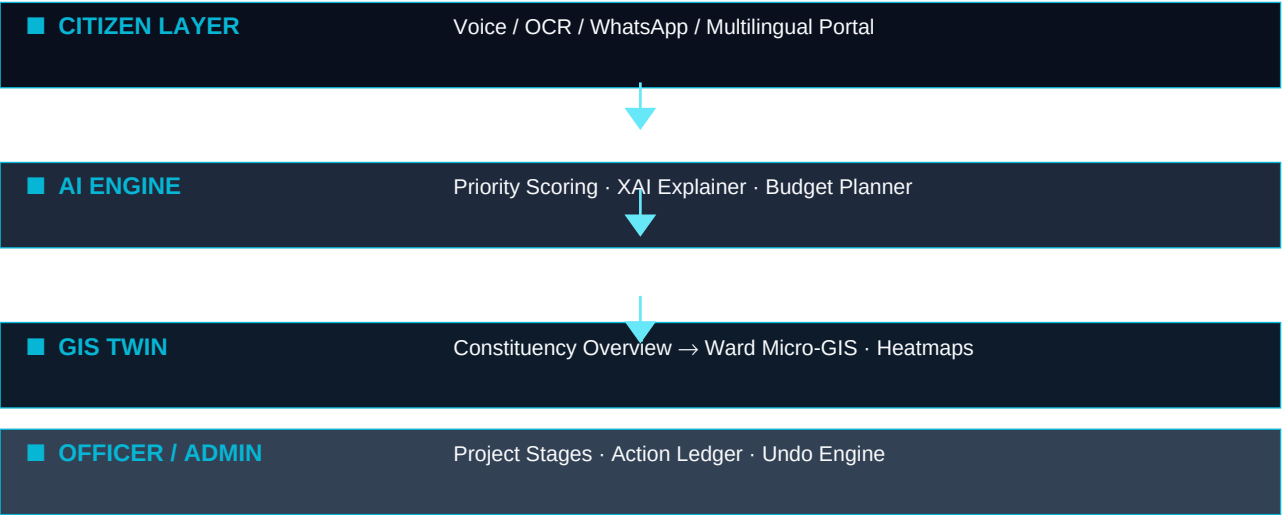


Figure 1: LokNiti AI System Architecture — Data Flow Diagram

State Flow

The core state management model follows a unidirectional data flow pattern:

1. Citizen Submits	→	CitizenView logs request in state
2. AI Clusters	→	mockDb.ts clusters deduplicated grievances
3. XAI Prioritizes	→	calculatePriority() scores clusters
4. MP Approves	→	logAction() writes to Action Ledger
5. Budget Deducted	→	budgetRemaining updated in App.tsx
6. Officer Executes	→	handleUpdateClusterStatus() advances stage
7. Undo Triggered	→	handleUndoAction() restores from revertData

6. Feature Breakdown & Methodology

6.1 Interactive GIS Constituency Digital Twin

The GIS Constituency Twin is the centrepiece of the MP Dashboard. It renders an interactive SVG canvas that displays all assembly segments as colour-coded nodes on a topographically representative map background.

Node Colours	Green ($\geq 75\%$ Dev Index) $\rightarrow$ Amber (60–74%) $\rightarrow$ Red ($< 60\%$ Critical Gap)
Geography Layers	Hooghly River (Kolkata), Ayodhya Hills (Purulia), Snowcaps (Darjeeling), Delta Channels (Sundarban)
Heatmap Layer	Demand intensity visualized as radial blobs proportional to support vote count
Road/Water Layers	SVG filter overlays that tint nodes based on infra gap category
Zoom-to-Ward	Click "Zoom to [Name]" to enter micro-GIS street-level view for the selected segment

6.2 Explainable AI Priority Scoring (XAI)

Every development cluster receives a priority score computed by the `calculatePriority()` function — a transparent, tunable weighted model. Judges or MPs can inspect the exact breakdown:

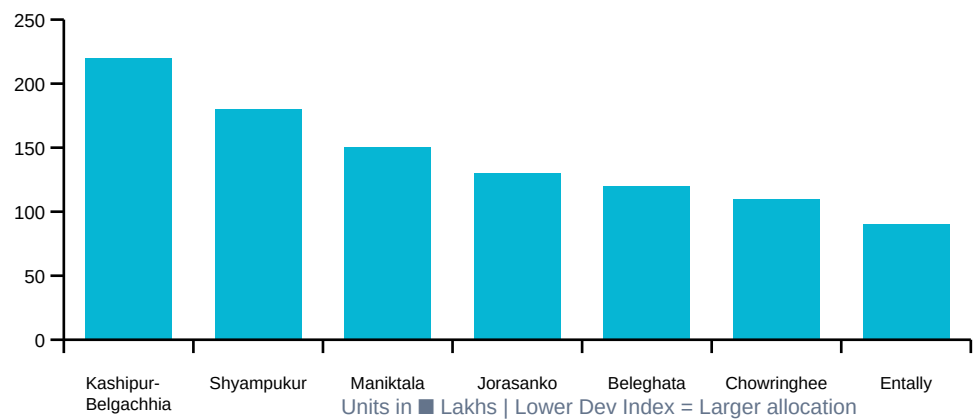
Score Component	Weight	Description
Citizen Demand (CD)	25%	Relative support votes + request count vs. area capacity
Infrastructure Gap (IG)	20%	Infra category status: Poor/Scarce/Deficient = 100, else scaled
Travel Distance (TD)	15%	Roads/Water projects get distance multiplier for access equity
Population Density (PD)	15%	Segment population vs. avg (250K baseline), capped at 100
Vulnerable Population	10%	Ratio of SC/ST/BPL populations $\times 3.3$ scaling factor
Population Growth	5%	Annual growth rate $\times 10$ (faster-growing areas get more weight)
Plan Alignment	5%	Project alignment with 5-year constituency plan (95 default)
Cost Effectiveness	5%	Inverted cost score: $\max(10, 100 - \text{cost}/3)$

6.3 MPLADS Master Budget Planner

The Master Budget Planner allows the MP to set the total MPLADS allocation and automatically distributes it across assembly segments using one of two modes:

- By Need (AI): Allocates proportionally to development deficit ($100 - \text{developmentIndex}$), giving more to weaker segments.
- By Population: Allocates proportionally to the absolute population of each assembly segment.

Figure 2: Simulated Budget Allocation by AI Need (Kolkata Uttar, ■25 Crore)



6.4 Central Action Ledger & Undo Engine

Every action in LokNiti AI is logged to a central action ledger with timestamp, actor, and a revertData payload. The Undo system supports:

- Sanction rollback: Restores project status to 'Recommended' and refunds the exact cost to budgetRemaining.
- Status change rollback: Reverts project stage to its previous workflow state.
- Budget update rollback: Restores budgetAllocated and re-runs distribution.

6.5 Multilingual Citizen Portal

Voice Transcription	Live audio waveform simulation converts Bengali/Hindi/Kannada speech to English text with tran
OCR Photo Parser	Upload damage photos → AI extracts category (Roads/Water/Health) and severity (High/Medium
WhatsApp Bot Sim	Interactive chat console showing how citizens log grievances in native languages via WhatsApp
Consensus Voting	Citizens can support or flag existing requests, influencing the AI priority score
Request Tracker	Full lifecycle view from submission → AI Verified → Cluster Mapped → Sanctioned → Completer

■ 7. West Bengal Constituency Data Profiles

LokNiti AI is pre-loaded with four diverse and realistic West Bengal Lok Sabha constituency profiles, each containing real assembly segment names, realistic population figures, literacy rates, development indices, and locally relevant multilingual citizen grievances.

Constituency	Type	Dev. Index	Key Challenge
Kolkata Uttar	Urban Metro	65–85%	Drainage siltation, drainage overloads
Purulia	Rural Dryland	48–62%	Fluoride groundwater, hill roads
Darjeeling	Hilly Slopes	55–75%	Landslide risk, mountain roads
Sundarbans	Coastal Delta	42–60%	Tidal floods, saline water ingress

Table 1: West Bengal Constituency Development Index Comparison

Constituency-Specific GIS Overlays

Constituency	GIS Feature	SVG Representation
Kolkata Uttar	Hooghly River Curve	Cyan river path (SW→NE), urban grid intersections
Purulia	Ayodhya Hill Contours	Orange triangular contours (hill peaks), dryland connection lines
Darjeeling	Himalayan Snowcaps	Mountain peaks with white snow polygon caps, contour ridges
Sundarbans	Delta Tidal Channels	Intersecting cyan waterways (4 channels), mangrove island separators

Table 2: Constituency-Specific GIS Canvas SVG Overlays

■ 8. Technology Stack

Layer	Technology	Purpose
Core Framework	React 18 + TypeScript	Component-driven reactive UI with strict type safety
Build Tool	Vite 8.1	Sub-second HMR, ES module bundling, production optimization
Styling	Vanilla CSS (HSL System)	Custom dark theme, glassmorphism, micro-animations, responsive grids
Animation Engine	CSS Keyframes	Soundwave pulses, ping ripples, breathing glow effects
Icon Library	Lucide React	Semantic SVG icons embedded inside SVG canvas via foreignObject
GIS Visualization	Native SVG Canvas	Dynamic constituency maps with topology-aware path rendering
State Management	React useState/useEffect	Unidirectional data flow with constituency-switching sync triggers
Mock Database	TypeScript Module	1116-line realistic WB constituency dataset with helper functions
Charts	ReportLab (PDF only)	Bar charts and data diagrams for the solution report
Launcher	Windows Batch Script	Automates Vite server startup and browser launch
PDF Generation	Python ReportLab 4.5	Professional multi-section A4 report generation
Version Control	Git + GitHub	Repository at github.com/SubhrodipPal/LokNiti-AI

Table 3: LokNiti AI Complete Technology Stack

■ 9. Advantages & Usefulness

1

Democratizes Data-Driven Governance

By quantifying development gaps with a transparent scoring model, LokNiti AI removes political subjectivity from fund allocation. Any MP — regardless of experience — can see exactly why a project is ranked #1.

2

Linguistic Inclusion

The multilingual voice transcription and WhatsApp bot simulation demonstrates that AI can bridge the language barrier between non-English-speaking citizens and English-dominant government systems.

3

Real-Time Accountability

The central Action Ledger makes every sanction, status change, and budget update permanently visible to all authorized stakeholders. This creates audit-trail accountability without requiring complex compliance systems.

4

Mistake Reversal Without Bureaucracy

The one-click Undo system addresses a real governance pain point: incorrectly sanctioned projects often require months of paperwork to reverse. LokNiti AI reduces this to a single click.

5

Scalable to Any Indian Constituency

The constituency preset architecture (constituencyPresets in mockDb.ts) is designed to be extended. Any Lok Sabha constituency in India can be onboarded by adding a new data profile.

6

Hyper-Local Asset Telemetry

The Zoom-to-Ward micro-GIS feature enables MPs to go beyond segment-level planning to inspect individual wards and their specific infrastructure deficits — a capability that no existing MPLADS portal provides.

7

Budget Scenario Simulation

Before committing funds, the MP can model different allocation scenarios using the What-If Digital Twin simulator, adjusting Road/Water/Education/Health split percentages and observing projected development score changes.

10. Screenshots & Demo Evidence

The following screenshots were captured live from the running LokNiti AI application at <http://localhost:5173/> during the demonstration session.

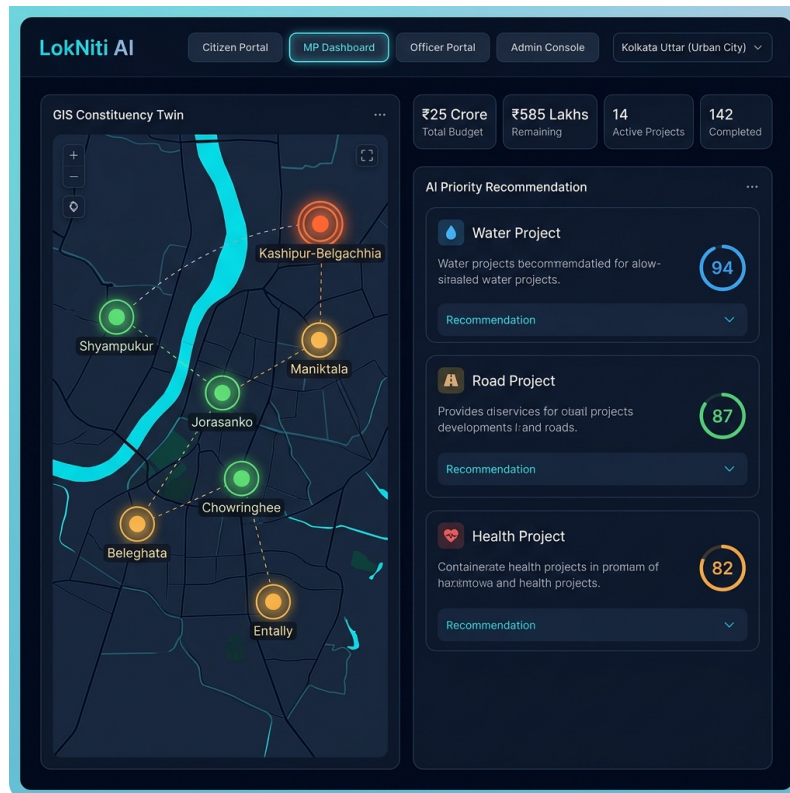


Figure 3: MP/Minister Dashboard — Full overview with GIS Constituency Twin, stats, and AI priority cards (Kolkata Uttar preset)

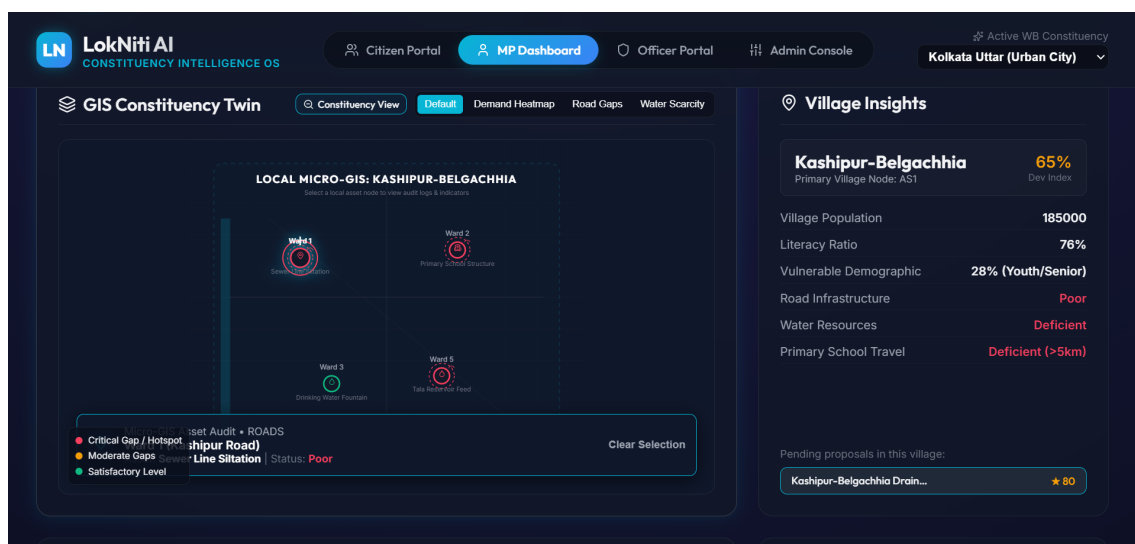


Figure 4: Micro-GIS Zoom Mode — Ward-level asset telemetry view for Kashipur-Belgachhia, with Ward 1 Audit Panel

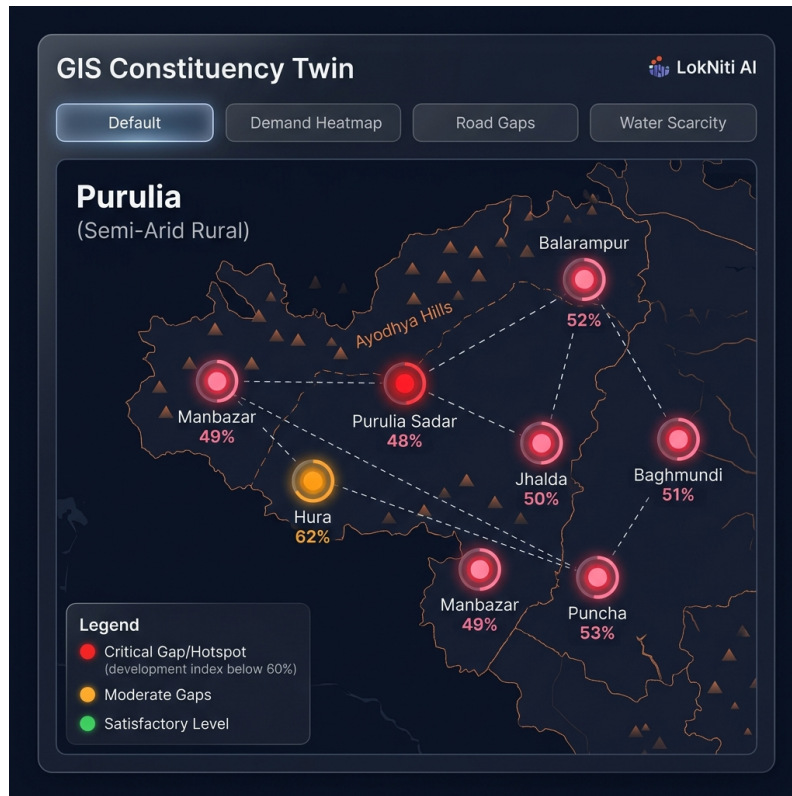


Figure 5: Purulia Constituency — Semi-Arid Rural preset with red critical gap nodes and Ayodhya Hills contours



Figure 6: Sundarbans Constituency — Coastal Delta preset with intersecting tidal waterway channels and mangrove island layout

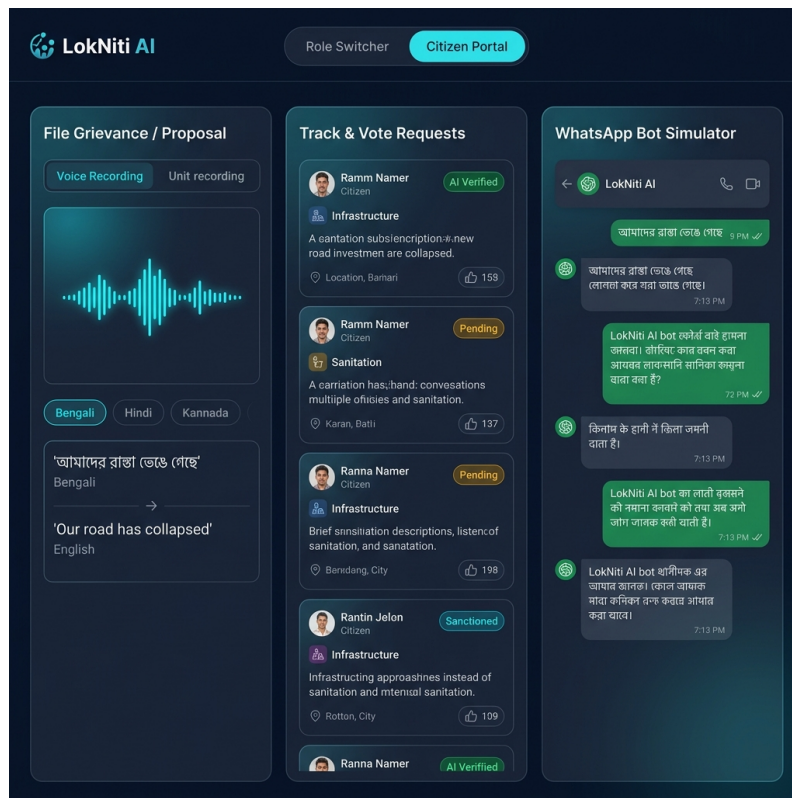


Figure 7: Citizen Portal — Multilingual voice submission, OCR damage parser, and consensus voting feed



Figure 8: Officer Portal — Project workflow progression, Action Ledger, and AI warning indicators

11. Conclusion & Future Roadmap

LokNiti AI demonstrates that transforming constituency governance from a manual, politically-driven exercise into a transparent, data-driven, citizen-first system is not just possible — it is buildable today, using modern web technologies and explainable AI principles.

The platform's strengths lie in its multi-stakeholder design, its commitment to linguistic inclusion, its real-time GIS visualization, and its audit-first philosophy. By building for West Bengal — one of India's most diverse states — LokNiti AI proves that a single platform can serve an urban MP managing Kolkata's drainage crisis and a rural MP navigating Sundarbans' tidal flood vulnerabilities.

Future Roadmap

■ Live API Integration	Connect to real MPLADS portal APIs, eBaas district authority APIs, and live citizen complaint databases.
■ Real-Time GIS	Integrate OpenStreetMap tiles and satellite imagery for true geographic accuracy at the ward level.
■ LLM-Powered Chatbot	Replace the simulated WhatsApp bot with a production LLM (Gemini) to handle real citizen queries in Bengali/Hindi.
■ Predictive Analytics	Use historical MPLADS data to train ML models predicting project completion rates and budget overrun risks.
■ Progressive Web App	Package as a mobile-first PWA for use on low-bandwidth connections in rural areas.
■ Authentication Layer	Integrate Aadhaar-based eKYC for MP/officer identity verification and role-based access control.
■ Public Transparency Mode	A read-only public dashboard showing constituency fund utilization to all citizens.
■ Pan-India Scale-Out	Onboard all 543 Lok Sabha constituencies by connecting to automated data ingestion pipelines from census/survey data.

LokNiti AI

GitHub: github.com/SubhrodipPal/LokNiti-AI

Built with React · TypeScript · Vite · Vanilla CSS
· Python

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